

BACHELOR WITH COMPUTER APPLICATIONS AS MAJOR (CT-3)

6th SEMESTER

CAP622J3: COMPUTER APPLICATIONS _ PROBABILITY AND STATISTICS

CREDITS: THEORY (4) PRACTICAL (2)

COURSE LEARNING OUTCOMES:

1. *Learn the language and core concepts of probability theory.*
2. *Understand basic principles of statistical inference.*
3. *Become an informed consumer of statistical information and have a good knowledge of what expectation and variance mean and be able to compute them.*

PREREQUISITE: Fundamental Mathematics

UNIT 1: (15 LECTURES)

Introduction to Probability: Random experiment, sample space, trial, event. Simple probability, Compound Probability, mutually exclusive events, Addition theorem, independent events, Multiplication theorem, Dependent events, Conditional probability, Bayes' Theorem, Partitions and Total probability law. Exploring Univariate Data: Types of data, Mean, Mode and Median.

UNIT 2: (15 LECTURES)

Standard Deviation and Variance, Range and Finding Outliers. Counting, Random variables, probability mass function, probability density function. distributions, quantiles, mean-variance, Joint distributions, covariance, correlation, independence, and Central limit theorem.

Discrete Distributions, Random Variables, Binomial Distributions, Geometric Distributions Continuous Distributions, Density Curves, The Normal Distribution

UNIT 3: (15 LECTURES)

Multivariate Data, Scatter Plots, Correlation, The Least Squares Regression Line, Residuals, Non-Linear Models, Relations in Categorical Data, Margins of Error and Estimates, Confidence Interval for a Proportion, Confidence Interval for the Difference of Two Proportions, Confidence Interval for a Mean, Confidence Interval for the Difference of Two Means.

UNIT 4: (15 LECTURES)

Tests of Significance, Inference for the Mean of a Population, Sample Proportions, Inference for a Population Proportion, Comparing Two Means, Comparing Two Proportions, Goodness of Fit Test, and Two-way Tables. Simple correlation (Pearson's correlation coefficient), Simple linear regression, Prediction, error in prediction, principle of least square.

TUTORIALS:

1. Two dice are rolled. Find the sample space for this experiment.
2. What is the probability of drawing a red card from a standard deck of 52 cards?
3. If two events are mutually exclusive, and the probability of either event A or event B occurring is 0.4, what is $P(A) + P(B)$?
4. If two events are independent, and the probability of event A is 0.3 and the probability of event B is 0.6, what is $P(A \text{ and } B)$?
5. If A and B are mutually exclusive events, and $P(A) = 0.2$, what is $P(B)$?
6. Calculate the conditional probability of event A given that event B has occurred if $P(A) = 0.4$ and $P(B) = 0.3$.
7. Using Bayes' Theorem, find the probability of event A if $P(B/A) = 0.6$, $P(A) = 0.4$, and $P(B) = 0.5$.
8. Define a random variable and provide an example of one.
9. If X is a discrete random variable with the following probability mass function: $P(X = 1) = 0.2$, $P(X = 2) = 0.4$, and $P(X = 3) = 0.4$, find $E(X)$, the expected value of X.
10. In a deck of cards, what is the probability of drawing a face card (jack, queen, or king)?
11. If you roll a fair six-sided die three times, what is the probability of getting three 6s in a row?
12. If there are 25 students in a class, and 5 of them wear glasses, what is the probability that a randomly selected student does not wear glasses?
13. If a bag contains 8 white balls and 6 black balls, what is the probability of drawing a white ball and then a black ball (without replacement)?
14. A box contains 12 chocolates, of which 3 are dark chocolate. What is the probability of randomly selecting a dark chocolate?
15. If you have a deck of cards and draw 2 cards with replacement, what is the probability that both cards are aces?

16. If you flip a coin three times, what is the probability of getting exactly two heads?
17. If a jar contains 30 red marbles and 20 blue marbles, what is the probability of drawing a red marble on the first try and a blue marble on the second try (without replacement)?
18. If you roll a fair six-sided die four times, what is the probability of getting at least one 1?
19. Imagine you are flipping a fair coin. How can you use probability distributions to represent the outcomes of this coin toss experiment?
20. Suppose you have a standard six-sided die. What is the probability mass function for this die, and what is the probability of rolling a 3?
21. You are conducting a survey where each respondent can either say "Yes" or "No" to a question. If you expect 70% of people to say "Yes," can you use the binomial distribution to calculate the probability of getting exactly 4 "Yes" responses out of 10 respondents?
22. You are modeling the outcome of a single basketball free throw attempt, where a player either makes it (success) or misses it (failure). Is this scenario best represented by a Bernoulli distribution? Why or why not?
23. Imagine you are tracking the number of customer arrivals at a small coffee shop per hour. Can you explain when and why you might use a Poisson distribution to model this situation?
24. Consider the heights of adult males in a population. Why is the normal distribution often used to describe this data, and what are the defining characteristics of the normal distribution?
25. If a student's test score has a z-score of -1.5, what does this mean in terms of their performance compared to the rest of the students? How can you use percentiles to describe their ranking within the class?
26. Suppose you have a population of 100 test scores with a mean of 75 and a standard deviation of 10.
10. If you take random samples of 30 test scores each and calculate the sample means, what is the expected mean of the sample means, and what is the standard error of the sample means? You're conducting a survey to estimate the average income of a population.
27. From a sample of 50 individuals, you find a sample mean income of 800000 and a sample standard deviation of 70000. Calculate a 95% confidence interval for the population mean income.
28. In a chi-square goodness-of-fit test, you expect a uniform distribution of colors in a bag of marbles, but you observe the following counts: Red (25), Blue (30), Green (20), and Yellow (25). Calculate the chi-square test statistic and determine if the observed distribution significantly differs from the expected uniform distribution at a 5% significance level
29. Suppose you have a dataset of 50 test scores, and the scores are normally distributed with an unknown mean (μ) and a known standard deviation (σ) of 15. If the maximum likelihood estimation gives you a mean of 65, what is the likelihood function for this dataset?
30. In a survey of 200 people, 120 said they prefer tea over coffee. Calculate a 95% confidence interval for the proportion of people in the entire population who prefer tea.

TEXTBOOK:

1. Probability and Statistics in Engineering (4th Edition) - W. Hines, D. Montgomery, D. Goldsman, C. Borror- Wiley Publication.
2. Introduction to Probability and Statistics for Engineers and Scientists (3rd Edition) - Sheldon M. Ross, Elsevier Academic Press.

REFERENCES:

1. Mood A.M. Graybill F.A and Boes D.C. (1974): Introduction to the Theory of Statistics. McGraw Hill.
2. Snedecor G.W and Cochran W.G. (1967); Statistical Methods. Iowa State University Press.
3. Cooke, Cramer and Clarke (1996): Basis Statistical Computing, Chapman and Hall. 4. David S. (1996): Elementary Probability, Oxford House.
4. Meyer P.L (1970): Introductory Probability and Statistical application, Addison Wesley.